TRAIN BOOKING SYSTEM

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I would like to express my sincere gratitude to our Principal SR.ROSMIN SABS and our Computer Science teacher for giving me the opportunity to do this project on **"TRAIN BOOKING SYSTEM"** using Python and MySQL.

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I would also like to thank my school for providing the necessary facilities and learning environment to complete this project successfully.

Finally, I express my heartfelt thanks to my parents and friends for their constant support, motivation, and encouragement during the preparation of this project.

This project has helped me improve my programming skills and gain practical knowledge of Python, MySQL, and database connectivity.

INTRODUCTION

The project is designed with a simple and user-friendly interface so that users can operate the system without difficulty. The menu-driven structure allows users to select different options according to their needs. Each operation is performed based on the user's selection, making the system easy to understand and use.

The system uses Python to perform various operations and MySQL to store and manage data. Python communicates with the MySQL database through the MySQL Connector module. This connection allows the program to insert new records, retrieve existing information, update booking details, and delete records when necessary.

Another important feature of this project is the management of passenger and booking records. The system stores essential details such as passenger name, age, train number, source, destination, journey date, seat number, and PNR number. These details help users access their booking information whenever required.

The ticket cancellation feature allows users to cancel their bookings when necessary. After a successful cancellation, the system updates the booking records and makes the cancelled seat available for future bookings. This helps maintain accurate information about seat availability in the database.

The project also helps reduce common errors that may occur during manual booking management. By storing information in a structured database, the system makes it easier to maintain records and retrieve the required information. Appropriate conditions and validations can also be used to handle invalid inputs and prevent incorrect operations.

This project provides practical experience in developing database-driven applications. It combines important concepts of computer science, including Python functions, conditional statements, loops, exception handling, SQL queries, database connectivity, and data management. These concepts are useful for developing larger applications in the future.

In conclusion, the Train Booking System is an educational project that demonstrates how programming and database management can be combined to create a useful application. It provides a basic understanding of railway reservation operations and serves as a foundation for developing more advanced booking systems with additional features such as online payments, real-time train tracking, and automatic ticket confirmation.

OBJECTIVES OF THE PROJECT

The main objective of this project is to apply programming and database concepts to a real-world problem. The project demonstrates how Python can be connected with MySQL to create a useful database application.

The objectives of the project are:

- To develop a simple and user-friendly Train Booking System.
- To store and manage train details using MySQL.
- To store passenger and ticket booking information.
- To provide user registration and login facilities.
- To allow users to search for trains according to source and destination.
- To book train tickets electronically.
- To automatically generate a unique PNR number.
- To assign seats to passengers.
- To update available seats after booking and cancellation.
- To provide a facility for cancelling booked tickets.
- To retrieve and display booking details easily.
- To reduce manual paperwork and human errors.
- To demonstrate the practical application of Python and MySQL.

PURPOSE

The Train Booking System is designed to provide a simple solution for managing train reservations and passenger information. In a manual railway booking system, maintaining passenger records, train details and ticket information can require considerable time and paperwork.This project stores the required information in a MySQL database. The user can register and log in to the system and then access different facilities through a menu-driven interface.The system provides the following facilities:

- User registration
- User login
- Train details
- Train search
- Ticket booking
- Ticket cancellation
- Booking details

When a ticket is booked, the system generates a PNR number and assigns a seat number. The available seats of the selected train are automatically reduced. When a ticket is cancelled, the available seat count is increased again.The project demonstrates how database managementcan be used to make a real-world reservation system more organized, efficient and easy to operate.

PYTHON

Python is a high-level, interpreted, general-purpose programming language. It was created by Guido van Rossum and was first released in 1991.

Python is widely used because of its simple syntax and readability. It supports different programming concepts and provides many built-in modules.

In this project, Python is used as the main programming language. The `mysql.connector` module is used to establish a connection between Python and the MySQL database.

The `random` module is used to generate a random number for creating the PNR number.

FEATURES

1. Free and easy to download and install.
2. Supports multiple programming paradigms.
3. Dynamically typed language.
4. Supports automatic memory management.
5. Provides exception handling.
6. Supports functions and code reusability.
7. Can be integrated with other programming languages.
8. Supports file handling.
9. Provides powerful libraries for data analysis and artificial intelligence.
10. Useful for web development and automation.
11. Supports GUI (Graphical User Interface) development.
12. Has a large community and good documentation.

MYSQL

MySQL is an open-source Relational Database Management System (RDBMS) that uses Structured Query Language (SQL). It is used to create, store, manage and retrieve data from databases.In this project, MySQL is used to store user information, train information and booking information. The database used in this project is named `register`.

The project contains the following tables:

- `users`
- `trains`
- `bookings`

These tables store the information required by the Train Booking System.

FEATURES

- It is open source and freely available.
- It is easy to use.
- It can store a large amount of data.
- It provides fast data retrieval.
- It supports relational database operations.
- It can be connected with Python using MySQL Connector.
- It provides facilities for inserting, updating, deleting and retrieving records.

HARDWARE AND SOFTWARE CONFIGURATION

HARDWARE

- Processor: Intel
- RAM: 4 GB or above
- Hard Disk: 80 GB or above

SOFTWARE

- Python
- MySQL
- MySQL Connector for Python
- Windows / Linux operating system

MODULES OF THE PROJECT

The Train Booking System consists of the following major modules:

1. USER REGISTRATION

New users can create an account by entering a username and password. The information is stored in the `users` table.

2. USER LOGIN

Registered users can enter their username and password. The system checks the details in the database before allowing access.

3. TRAIN DETAILS

The system displays the available trains along with train number, train name, source, destination, departure time, arrival time, available seats and fare.

4. SEARCH TRAINS

Users can enter the source and destination to search for matching trains.

5. BOOK TICKET

Users can enter passenger details and select a train. If seats are available, the system generates a PNR number, assigns a seat and stores the booking details.

6. CANCEL TICKET

A ticket can be cancelled by entering its PNR number. The booking is removed from the database and the available seat count is increased.

7. VIEW BOOKINGS

The system displays all the booked ticket details stored in the database.

SCREENSHOTS

OUTPUT

USER REGISTRATION

```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

============== RESTART: D:\python codes\booking system.py ==============
TRAIN BOOKING SYSTEM
..............


USER LOGIN
1:ENTER 1 FOR REGISTER
2: ENTER 2 FOR LOGIN
ENTER:1
USERNAME:sanjay
PASSWORD:1234
REGISTERED SUCCESSFULLY
```

USER LOGIN

```
=================== RESTART: D:\python codes\booking system.py ===================
TRAIN BOOKING SYSTEM
..............


USER LOGIN
1:ENTER 1 FOR REGISTER
2: ENTER 2 FOR LOGIN
ENTER:2
USERNAME:sanjay
PASSWORD:1234
LOGINED!

TRAIN BOOKING SYSTEM
--------------------
1: TRAIN DETAILS
2: SEARCH TRAINS
3: BOOK TICKET
4: CANCEL TICKET
5: VIEW BOOKINGS
6: EXIT
ENTER YOUR CHOICE:
```

TICKET BOOKING

```
BOOK TICKET
-----------
PASSENGER NAME:sanjay
AGE:18
GENDER:m
TRAIN NUMBER:12625
BOOKING DATE:07-10-2026

TICKET BOOKED SUCCESSFULLY
PNR: PNR97190
PASSENGER NAME: sanjay
TRAIN NUMBER: 12625
SEAT NUMBER: 1
FARE: 500
```

TICKET CANCELLATION

```
TRAIN BOOKING SYSTEM
--------------------
1: TRAIN DETAILS
2: SEARCH TRAINS
3: BOOK TICKET
4: CANCEL TICKET
5: VIEW BOOKINGS
6: EXIT
ENTER YOUR CHOICE:4
ENTER PNR:PNR54321
BOOKING CANCELLED
AVAILABLE SEAT INCREASED

TRAIN BOOKING SYSTEM
```

ALL BOOKED TICKETS

```
ALL BOOKED TICKETS
------------------
PNR: PNR97190
PASSENGER NAME: sanjay
AGE: 18
GENDER: m
TRAIN NUMBER: 12625
SEAT NUMBER: 1
FARE: 500
BOOKING DATE: 07-10-2026
--------------------
```

TRAIN DETAILS

```
TRAIN DETAILS
-------------
TRAIN NUMBER: 12625
TRAIN NAME: Kerala Express
SOURCE: Kottayam
DESTINATION: New Delhi
TIME: 10:30 AM
FARE: 500
AVAILABLE SEATS: 50
--------------------
TRAIN NUMBER: 12626
TRAIN NAME: Malabar Express
SOURCE: Kochi
DESTINATION: Chennai
TIME: 08:00 AM
FARE: 450
AVAILABLE SEATS: 50
--------------------
TRAIN NUMBER: 12627
TRAIN NAME: Venad Express
SOURCE: Kollam
DESTINATION: Ernakulam
TIME: 07:00 AM
FARE: 200
AVAILABLE SEATS: 50
--------------------
```

TABLES CREATED

```
mysql> use register;
Database changed
mysql> show tables;
+--------------------+
| Tables_in_register |
+--------------------+
| passengers         |
| trains             |
| users              |
+--------------------+
3 rows in set (0.257 sec)

mysql> desc users;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| USERNAME | varchar(50) | YES  |     | NULL    |       |
| PASSWORD | varchar(50) | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
2 rows in set (0.112 sec)

mysql> desc trains;
+-----------------+-------------+------+-----+---------+-------+
| Field           | Type        | Null | Key | Default | Extra |
+-----------------+-------------+------+-----+---------+-------+
| TRAIN_NO        | int         | YES  |     | NULL    |       |
| TRAIN_NAME      | varchar(50) | YES  |     | NULL    |       |
| SOURCE          | varchar(50) | YES  |     | NULL    |       |
| DEST            | varchar(50) | YES  |     | NULL    |       |
| TIME            | varchar(50) | YES  |     | NULL    |       |
| FARE            | int         | YES  |     | NULL    |       |
| AVAILABLE_SEATS | int         | YES  |     | NULL    |       |
+-----------------+-------------+------+-----+---------+-------+
7 rows in set (0.027 sec)

mysql> desc passengers;
+--------------+-------------+------+-----+---------+-------+
| Field        | Type        | Null | Key | Default | Extra |
+--------------+-------------+------+-----+---------+-------+
| NAME         | varchar(50) | YES  |     | NULL    |       |
| AGE          | int         | YES  |     | NULL    |       |
| GENDER       | varchar(1)  | YES  |     | NULL    |       |
| TRAIN_NO     | int         | YES  |     | NULL    |       |
| TRAIN_NAME   | varchar(50) | YES  |     | NULL    |       |
| BOOKING_DATE | varchar(50) | YES  |     | NULL    |       |
| PNR          | varchar(20) | YES  |     | NULL    |       |
| SEAT_NO      | int         | YES  |     | NULL    |       |
+--------------+-------------+------+-----+---------+-------+
8 rows in set (0.031 sec)

mysql>
```

```
mysql> desc bookings;
+----------------+-------------+------+-----+---------+-------+
| Field          | Type        | Null | Key | Default | Extra |
+----------------+-------------+------+-----+---------+-------+
| PNR            | varchar(20) | YES  |     | NULL    |       |
| PASSENGER_NAME | varchar(50) | YES  |     | NULL    |       |
| AGE            | int         | YES  |     | NULL    |       |
| GENDER         | varchar(1)  | YES  |     | NULL    |       |
| TRAIN_NO       | int         | YES  |     | NULL    |       |
| SEAT_NO        | int         | YES  |     | NULL    |       |
| FARE           | int         | YES  |     | NULL    |       |
| BOOKING_DATE   | varchar(50) | YES  |     | NULL    |       |
+----------------+-------------+------+-----+---------+-------+
8 rows in set (0.562 sec)
```

SOURCE CODE OF TRAIN BOOKING SYSTEM

```
import mysql.connector as tor
import random

print("TRAIN BOOKING SYSTEM")
print(".....")
print()
print("USER LOGIN")
print("""1: ENTER 1 FOR REGISTER
2: ENTER 2 FOR LOGIN""")

a=int(input("ENTER:"))

if a==1:
   my=tor.connect(host='localhost',user='root',password='12345678',database='register')
   h=my.cursor()
   u=input("USERNAME:")
   p=input("PASSWORD:")
   q="insert into users(USERNAME,PASSWORD) values('{}','{}')".format(u,p)
   h.execute(q)
   my.commit()
   print("REGISTERED SUCCESSFULLY")
   my.close()

elif a==2:
   my=tor.connect(host='localhost',user='root',password='12345678',database='register')
   h=my.cursor()

   u=input("USERNAME:")
   p=input("PASSWORD:")

   k="select * from users where USERNAME='{}' and PASSWORD='{}'".format(u,p)
   h.execute(k)
   result=h.fetchone()

   if result:
      print("LOGINED!")

      while True:
         print()
         print("TRAIN BOOKING SYSTEM")
         print("--------------------")
         print("1: TRAIN DETAILS")
         print("2: SEARCH TRAINS")
         print("3: BOOK TICKET")
         print("4: CANCEL TICKET")
```

```
        print("5: VIEW BOOKINGS")
        print("6: EXIT")

        ch=int(input("ENTER YOUR CHOICE:"))

        if ch==1:

my=tor.connect(host='localhost',user='root',password='12345678',database='register')
            h=my.cursor()

            h.execute("select * from trains")
            data=h.fetchall()

            print()
            print("TRAIN DETAILS")
            print("-------------")

            for x in data:
               print("TRAIN NUMBER:",x[0])
               print("TRAIN NAME:",x[1])
               print("SOURCE:",x[2])
               print("DESTINATION:",x[3])
               print("TIME:",x[4])
               print("FARE:",x[5])
               print("AVAILABLE SEATS:",x[6])
               print("--------------------")

            my.close()

        elif ch==2:

my=tor.connect(host='localhost',user='root',password='12345678',database='register')
            h=my.cursor()

            s=input("SOURCE:")
            d=input("DESTINATION:")

            q="select * from trains where SOURCE='{}' and DEST='{}'".format(s,d)
            h.execute(q)
            data=h.fetchall()

            if data:
               print()
               print("MATCHING TRAINS")
               print("----------------")

               for x in data:
                  print("TRAIN NUMBER:",x[0])
                  print("TRAIN NAME:",x[1])
                  print("SOURCE:",x[2])
```

```
                print("DESTINATION:",x[3])
                print("TIME:",x[4])
                print("FARE:",x[5])
                print("AVAILABLE SEATS:",x[6])
                print("--------------------")
            else:
               print("NO TRAIN FOUND!")

            my.close()

        elif ch==3:

my=tor.connect(host='localhost',user='root',password='12345678',database='register')
            h=my.cursor()

            print()
            print("BOOK TICKET")
            print("-----------")

            n=input("PASSENGER NAME:")
            age=int(input("AGE:"))
            g=input("GENDER:")
            tn=int(input("TRAIN NUMBER:"))
            bd=input("BOOKING DATE:")

            q="select * from trains where TRAIN_NO={}".format(tn)
            h.execute(q)
            train=h.fetchone()

            if train:
               seats=train[6]
               fare=train[5]

               if seats>0:
                  seat=50-seats+1
                  pnr="PNR"+str(random.randint(10000,99999))

                  q="insert into
bookings(PNR,PASSENGER_NAME,AGE,GENDER,TRAIN_NO,SEAT_NO,FARE,BOO
KING_DATE) values('{}','{}',{},'{}',{},{},{},'{}')".format(pnr,n,age,g,tn,seat,fare,bd)

                  h.execute(q)

                  q="update trains set AVAILABLE_SEATS=AVAILABLE_SEATS-1 where
TRAIN_NO={}".format(tn)
                  h.execute(q)

                  my.commit()

                  print()
```

```
                print("TICKET BOOKED SUCCESSFULLY")
                print("PNR:",pnr)
                print("PASSENGER NAME:",n)
                print("TRAIN NUMBER:",tn)
                print("SEAT NUMBER:",seat)
                print("FARE:",fare)

            else:
                print("NO AVAILABLE SEATS!")

        else:
            print("TRAIN NOT FOUND!")

        my.close()

    elif ch==4:

my=tor.connect(host='localhost',user='root',password='12345678',database='register')
        h=my.cursor()

        pnr=input("ENTER PNR:")

        q="select TRAIN_NO from bookings where PNR='{}'".format(pnr)
        h.execute(q)
        data=h.fetchone()

        if data:
            tn=data[0]

            q="delete from bookings where PNR='{}'".format(pnr)
            h.execute(q)

            q="update trains set AVAILABLE_SEATS=AVAILABLE_SEATS+1 where
TRAIN_NO={}".format(tn)
            h.execute(q)

            my.commit()

            print("BOOKING CANCELLED")
            print("AVAILABLE SEAT INCREASED")

        else:
            print("INVALID PNR!")

        my.close()

    elif ch==5:

my=tor.connect(host='localhost',user='root',password='12345678',database='register')
        h=my.cursor()
```

```
            h.execute("select * from bookings")
            data=h.fetchall()

            print()
            print("ALL BOOKED TICKETS")
            print("------------------")

            if data:
               for x in data:
                  print("PNR:",x[0])
                  print("PASSENGER NAME:",x[1])
                  print("AGE:",x[2])
                  print("GENDER:",x[3])
                  print("TRAIN NUMBER:",x[4])
                  print("SEAT NUMBER:",x[5])
                  print("FARE:",x[6])
                  print("BOOKING DATE:",x[7])
                  print("--------------------")
            else:
               print("NO BOOKINGS FOUND!")

            my.close()

         elif ch==6:
            print("THANK YOU FOR USING TRAIN BOOKING SYSTEM")
            break

         else:
            print("WRONG CHOICE!")

      my.close()

   else:
      print("CHECK YOUR USERNAME AND PASSWORD!")

else:
   print("YOU HAVE ENTERED A WRONG CHOICE")
```

SOURCE CODE FOR TABLE CREATION

```
import mysql.connector as tor

my=tor.connect(host='localhost',user='root',password='12345678')
h=my.cursor()

h.execute("create database if not exists register")
h.execute("use register")

h.execute("""create table if not exists users(
USERNAME varchar(50),
PASSWORD varchar(50))""")

h.execute("""create table if not exists trains(
TRAIN_NO int,
TRAIN_NAME varchar(50),
SOURCE varchar(50),
DEST varchar(50),
TIME varchar(50),
FARE int,
AVAILABLE_SEATS int)""")

h.execute("""create table if not exists passengers(
NAME varchar(50),
AGE int,
GENDER varchar(1),
TRAIN_NO int,
TRAIN_NAME varchar(50),
BOOKING_DATE varchar(50),
PNR varchar(20),
SEAT_NO int)""")

h.execute("""create table if not exists bookings(
PNR varchar(20),
PASSENGER_NAME varchar(50),
AGE int,
GENDER varchar(1),
TRAIN_NO int,
SEAT_NO int,
FARE int,
BOOKING_DATE varchar(50))""")

my.commit()

print("TABLES CREATED SUCCESSFULLY")

my.close()
```

CONCLUSION

The Train Booking System is a database-based application developed using Python and MySQL. The project provides a simple and efficient method for managing train details, user accounts, passenger information and ticket bookings.

The system allows users to register and log in securely, search for trains, book tickets, cancel tickets and view booking information. It also automatically generates a PNR number and updates the available seats whenever a ticket is booked or cancelled.

This project helped in understanding the practical application of Python programming and MySQL database management. It also demonstrates how programming can be used to solve real-world problems and reduce manual work.

Overall, the Train Booking System provides a simple, organized and efficient method of managing railway ticket reservations.

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